Increments, and the problem of the tangent

Two questions from opposite ends of science — the speed of a falling stone and the slope of a curve — that turn out to be the same question.

LESSON 1–2 | 2 × 40 MIN | ALGEBRA 11, §1.1 | P1 · 7.1

LESSON CLOCK

8' | 16' | 18' | 20' | 20' | 8'

Two problems, one answer	8 min
Δx and Δy	16 min
The difference quotient	18 min
Secant to tangent	20 min
Practice	20 min
Homework	8 min

LEARNING OBJECTIVES

- Define the increment of the argument and the increment of the function.
- Compute Δy for a given function and Δx.
- Form the difference quotient and interpret it as an average rate of change.
- Explain why a tangent cannot be found by the two-point method alone.

TEXTBOOK

National	pp. 5–14
Cambridge	Pure Mathematics 1, pp. 122–126
Workbook	P1 Exercise 7A

01 Explanation

Two old problems

The physicist's problem. A stone falls $s = 5t^2$ metres in t seconds. Its average speed over the first 2 seconds is $20 \div 2 = 10$ m/s. But how fast is it moving **at the instant** $t = 2$? Dividing distance by time needs an interval, and an instant has none.

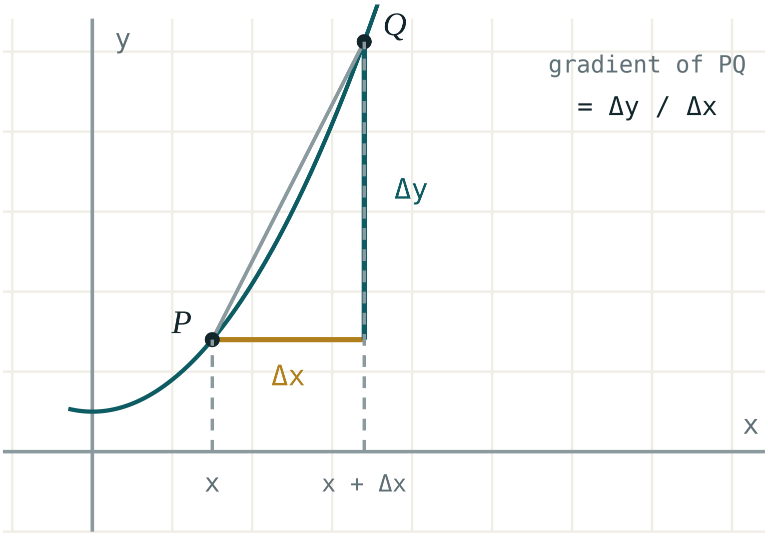
The geometer's problem. The gradient of a straight line is $\frac{\text{rise}}{\text{run}}$ between any two of its points. A curve has a different steepness everywhere. What is the gradient **at one point**?

> THE SAME QUESTION
>
> Both ask for a ratio at a single point, where the ratio has nothing to divide. Both are answered by the same idea: compute the ratio over a small interval, then shrink the interval.

Increments

Move the input from x_0 to a nearby x. The change in the input is the **increment of the argument**; the change it causes in the output is the **increment of the function**:

$$\Delta x = x - x_0 \quad \Delta y = f(x_0 + \Delta x) - f(x_0)$$



Δx along the axis, Δy up the side. The secant is the hypotenuse of that triangle.

Δx IS ONE SYMBOL

It is not Δ multiplied by x. It cannot be cancelled, split, or treated as a product. It may be positive or negative — the point may move left.

For $y = x^2$ at $x_0 = 3$ with $\Delta x = 0.1$: $\Delta y = 3.1^2 - 3^2 = 9.61 - 9 = 0.61$.

The difference quotient

Their ratio is the **average rate of change** over the interval — and geometrically it is the gradient of the **secant** through the two points:

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

For $y = x^2$ at $x_0 = 3$, expand once and the pattern appears:

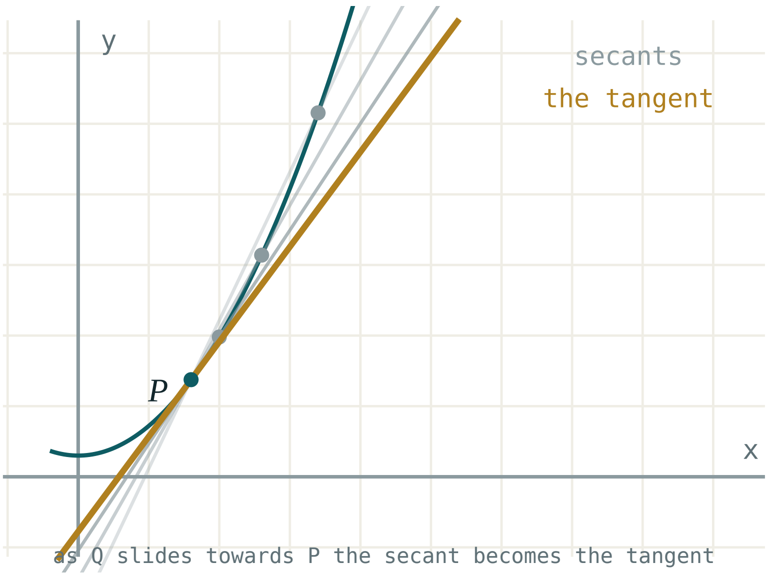
$$\frac{(3 + \Delta x)^2 - 9}{\Delta x} = \frac{6\Delta x + (\Delta x)^2}{\Delta x} = 6 + \Delta x$$

ΔX	ΔY	$\Delta Y/\Delta X$
1	7	7
0.1	0.61	6.1
0.01	0.0601	6.01
0.001	0.006001	6.001

The quotient is walking towards 6 and never arrives. That number is the answer to both problems.

Secant to tangent

Hold A fixed and slide B towards it along the curve. The secant AB turns, and settles into a limiting position: the **tangent** at A.



Three secants and the line they are approaching. The tangent is the limit, not one of them.

YOU CANNOT JUST SET $\Delta X = 0$

That gives $\frac{0}{0}$, which is meaningless. The whole of the next lesson exists to make “approaches without reaching” into something precise.

02 Worked examples

EXAMPLE 1 **For $y = x^2$, find Δy at $x_0 = 4$ with $\Delta x = 0.2$.**

1. $f(4.2) = 17.64$
2. $f(4) = 16$
3. $\Delta y = 1.64$
4. $\dfrac{\Delta y}{\Delta x} = \dfrac{1.64}{0.2} = 8.2$

   Close to 8.

ANSWER

$\Delta y = 1.64, \dfrac{\Delta y}{\Delta x} = 8.2$

EXAMPLE 2 **Simplify $\dfrac{\Delta y}{\Delta x}$ for $y = 3x^2 - x$ at a general x.**

1. $\Delta y = 3(x + \Delta x)^2 - (x + \Delta x) - (3x^2 - x)$
2. $= 6x\Delta x + 3(\Delta x)^2 - \Delta x$

   The $3x^2$ and x terms cancel.
3. $\dfrac{\Delta y}{\Delta x} = 6x + 3\Delta x - 1$

ANSWER

$6x + 3\Delta x - 1$

EXAMPLE 3 **A stone falls $s = 5t^2$. Find its average speed on $[2, 2.01]$.**

1. $s(2.01) = 5 \times 4.0401 = 20.2005$
2. $s(2) = 20$
3. $\Delta s = 0.2005$, $\Delta t = 0.01$
4. 20.05 m/s.

   Approaching 20.

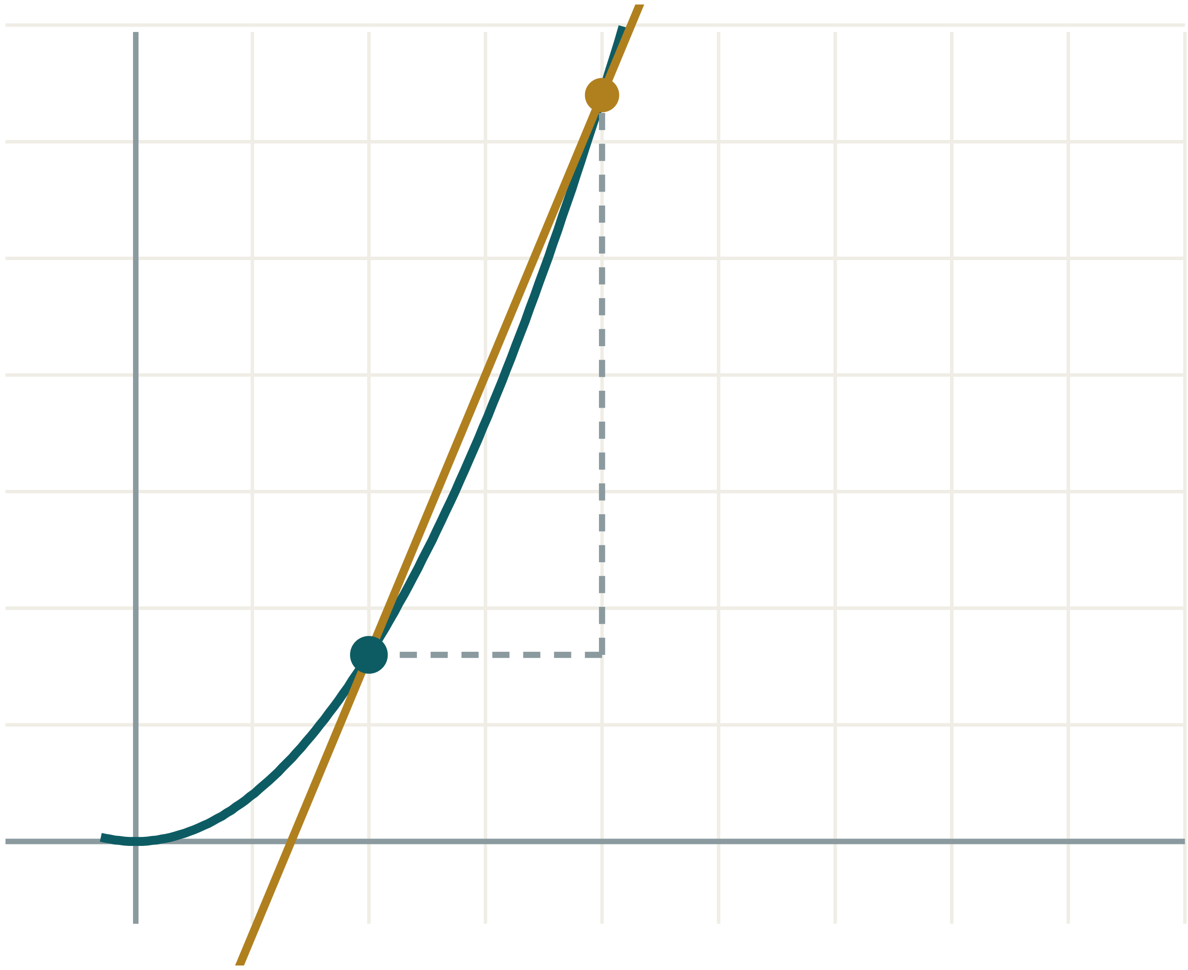
ANSWER

20.05 m/s

03 Interactive model

Slide h towards zero and read the gradient of the secant as it settles.

Secant to tangent



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Increment of the argument	Argument orttirmasi	Приращение аргумента
Increment of the function	Funksiya orttirmasi	Приращение функции
Difference quotient	Ayirmalar nisbati	Разностное отношение
Average rate of change	O‘rtacha o‘zgarish tezligi	Средняя скорость изменения
Instantaneous rate	Oniy tezlik	Мгновенная скорость
Secant	Kesuvchi	Секущая
Tangent	Urinma	Касательная
Gradient	Burchak koeffitsienti	Угловой коэффициент
Limiting position	Limit holat	Предельное положение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Δx means:

- Δ times x
- the change in x
- the derivative
- a small positive number

$\frac{\Delta y}{\Delta x}$ is the gradient of:

- the tangent
- the secant
- the normal
- the axis

For $y = x^2$ at $x = 3$, $\frac{\Delta y}{\Delta x}$ simplifies to:

- 6
- $6 + \Delta x$
- $3 + \Delta x$
- 9

Setting $\Delta x = 0$ directly gives:

- the answer
- $\frac{0}{0}$
- zero
- infinity

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $y = x^2$, $x_0 = 2$, $\Delta x = 1$. Find Δy

   ANSWER 5

2. $y = x^2$, $x_0 = 2$, $\Delta x = 0.1$. Find Δy

   ANSWER 0.41

3. $y = 3x$, any x_0, $\Delta x = 2$. Find Δy

   ANSWER 6

4. $y = 5$. Find Δy

   ANSWER 0

5. $y = x$. Find $\frac{\Delta y}{\Delta x}$

   ANSWER 1

6. $s = 5t^2$. Average speed on $[1, 2]$

   ANSWER 15 m/s

7. What does the secant become as $\Delta x \to 0$?

   ANSWER the tangent

Medium · the standard question

7 PROBLEMS

1. $y = x^2$. Simplify $\dfrac{\Delta y}{\Delta x}$

   ANSWER $2x + \Delta x$

2. $y = x^3$. Simplify $\dfrac{\Delta y}{\Delta x}$

   ANSWER $3x^2 + 3x\Delta x + (\Delta x)^2$

3. $y = 2x^2 + 1$. Simplify $\dfrac{\Delta y}{\Delta x}$

   ANSWER $4x + 2\Delta x$

4. $y = \dfrac{1}{x}$. Simplify $\dfrac{\Delta y}{\Delta x}$

   ANSWER $-\dfrac{1}{x(x + \Delta x)}$

5. $s = 5t^2$. Average speed on $[2, 2.1]$

   ANSWER 20.5 m/s

6. $y = x^2 - 4x$. $\dfrac{\Delta y}{\Delta x}$ at $x = 1$

   ANSWER $-2 + \Delta x$

7. What number is $2x + \Delta x$ approaching?

   ANSWER $2x$

Hard · stretch and reason

7 PROBLEMS

1. $y = \sqrt{x}$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $\frac{1}{\sqrt{x + \Delta x} + \sqrt{x}}$

2. $y = \frac{1}{x^2}$. Simplify $\frac{\Delta y}{\Delta x}$

ANSWER $-\frac{2x + \Delta x}{x^2(x + \Delta x)^2}$

3. Explain why $\frac{\Delta y}{\Delta x}$ cannot be evaluated at $\Delta x = 0$

ANSWER Both parts vanish; $\frac{0}{0}$ is undefined

4. $y = x^2$. For which Δx is the secant gradient at $x = 3$ equal to 6.5?

ANSWER $\Delta x = 0.5$

5. A ball: $h = 20t - 5t^2$. Average velocity on $[1, 1 + \Delta x]$

ANSWER $10 - 5\Delta x$

6. When is that average velocity zero?

ANSWER $\Delta x = 2$

7. Interpret the sign of $\frac{\Delta y}{\Delta x}$ for a decreasing function

ANSWER It is negative

07 Homework

Homework — 5 tasks

Task 5 needs a table, like the one in the lesson.

1. $y = x^2$. Find Δy at $x_0 = 5$ for $\Delta x = 1, 0.1, 0.01$.

2. $y = x^3 - x$. Simplify $\frac{\Delta y}{\Delta x}$ at a general x.
3. $s = 4t^2 + t$. Find the average speed on $[3, 3.1]$.
4. Explain in three sentences why the tangent cannot be found from two points on the curve.
5. Tabulate $\frac{\Delta y}{\Delta x}$ for $y = x^2$ at $x = 2$ with $\Delta x = 1, 0.5, 0.1, 0.01$ and say what it approaches.